

EXPERIMENT 7

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Project Phase I

Requirement Analysis and Database Design

Case study: EstateLink - Real Estate Brokerage Management

Aim	Select a case study, identify entities and relationships, prepare the ER diagram, convert it into a relational schema and normalize the database.

Contents

- 1. Case study and requirement analysis
- 2. Entities, attributes and relationships
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1. Case study and requirement analysis

EstateLink is a real estate brokerage that connects clients (buyers and tenants) with properties through brokers. Inquiries, property listings, commissions, leases and deals are tracked on paper or in spreadsheets, which causes duplicate records, lost follow-ups and no single view of a deal. The database must keep all of this in one consistent place.

Functional requirements

- Maintain broker records (name, contact details, joining date) and client records (contact details, buyer or tenant type).
- Maintain property listings with size, location, value, status, owner details and the tenant types the property is open to.
- Record commission rules (owner and client brokerage percentage) and link them to brokers and properties.
- Record every enquiry a client makes, the property it concerns and the broker who handles it.
- Record deals (sale or rental) with dates, value, status, rental period and the earnings of the broker and the company; a deal can come from an enquiry.
- Record lease details (type, security deposit, duration) for properties.
- Let clients write reviews (comments, rating, date) about their experience.

Business rules

- A broker has many properties, handles many enquiries, closes many deals and earns commission.
- A client submits many enquiries, enters many deals and writes many reviews.
- A property receives many enquiries, is involved in many deals and has leases.
- An enquiry can be converted into a deal.

2. Entities, attributes and relationships

Entities and their attributes (as drawn in the ER diagram)

BROKERS	broker_id, name, email, phone, join_date	broker_id
CLIENTS	client_id, name, email, phone, client_type	client_id
PROPERTIES	property_id, square_feet, owner_name, owner_contact, location, status, value, eligible_tenant_type	property_id
ENQUIRIES	enquiry_id, enquiry_date, status, notes	enquiry_id
DEALS	deal_id, deal_type, deal_date, deal_value, deal_status, rental_start_date, rental_end_date, broker_earning, company_earning	deal_id
COMMISSION	commission_id, owner_brokerage_percent, client_brokerage_percent	commission_id
LEASES	lease_id, lease_type, security_deposit, duration_months	lease_id
REVIEWS	review_id, comments, rating, review_date	review_id

In the sketch, phone is written as phno and lease_type is written as type.

Relationships

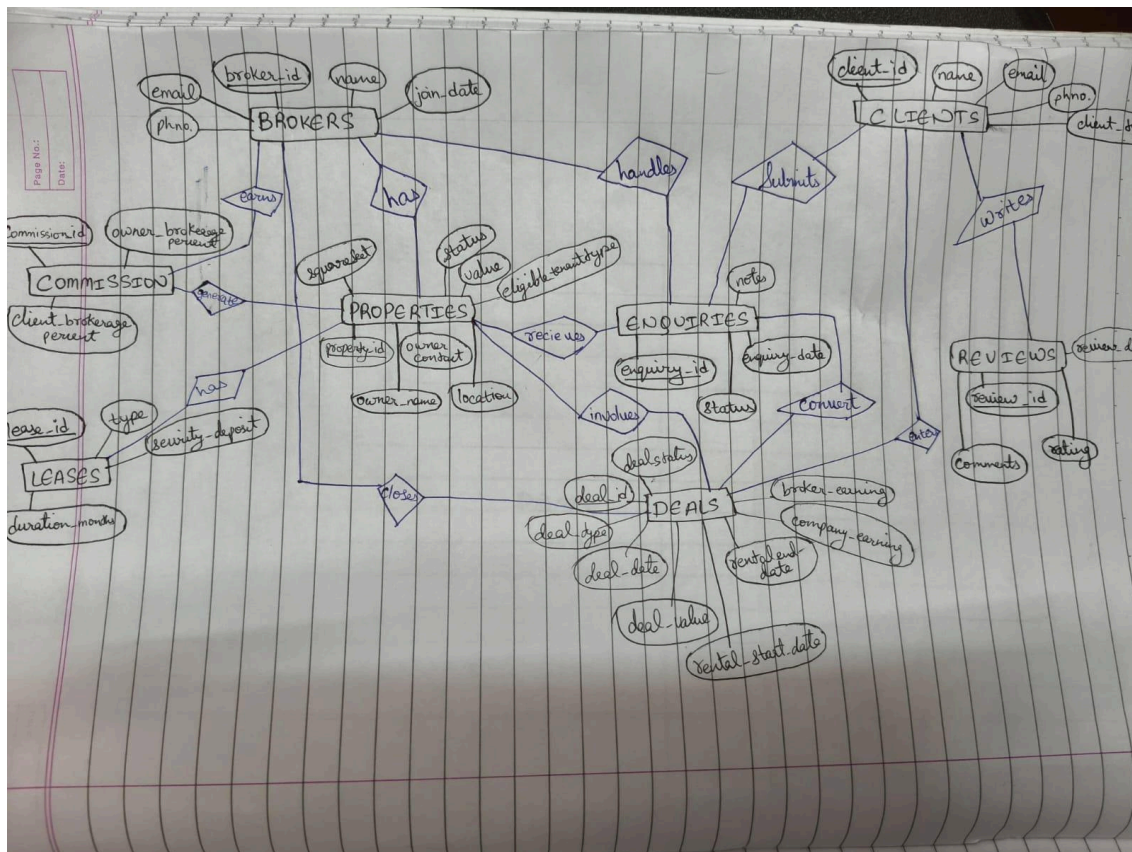
Every relationship connects exactly two entities. The sketch does not mark cardinality, so the usual one-to-many cardinality is assumed (enquiry-deal is one-to-one).

earns	BROKERS - COMMISSION	1 : N	A broker earns commission
generates	COMMISSION - PROPERTIES	1 : N	A commission rule applies to properties
has	BROKERS - PROPERTIES	1 : N	A broker has many properties
handles	BROKERS - ENQUIRIES	1 : N	A broker handles many enquiries
closes	BROKERS - DEALS	1 : N	A broker closes many deals
receives	PROPERTIES - ENQUIRIES	1 : N	A property receives many enquiries

involves	PROPERTIES - DEALS	1 : N	A property is involved in many deals
has	PROPERTIES - LEASES	1 : N	A property has leases
submits	CLIENTS - ENQUIRIES	1 : N	A client submits many enquiries
converts	ENQUIRIES - DEALS	1 : 1	An enquiry is converted into a deal
enters	CLIENTS - DEALS	1 : N	A client enters many deals
writes	CLIENTS - REVIEWS	1 : N	A client writes many reviews

3. Entity Relationship (ER) diagram

Notation: entities are rectangles, relationships are diamonds, attributes are ovals and primary keys are underlined.



4. Types of attributes

Each attribute in the ER diagram is classified below in one line, using these types:

Simple	Cannot be divided into smaller parts	status, deal_value
Composite	Made up of smaller sub-attributes	location (area, city, pincode)
Multivalued	Can hold more than one value for one entity instance	eligible_tenant_type (family, bachelor, student)

Stored	Value is kept in the database	join_date
Derived	Calculated from other attributes, not stored	broker_earning (from deal_value and brokerage percent)

Classification of every attribute

		Type
BROKERS	broker_id	Simple, Stored
BROKERS	name	Simple, Stored
BROKERS	email	Simple, Stored
BROKERS	phone	Simple, Stored
BROKERS	join_date	Simple, Stored
CLIENTS	client_id	Simple, Stored
CLIENTS	name	Simple, Stored
CLIENTS	email	Simple, Stored
CLIENTS	phone	Simple, Stored
CLIENTS	client_type	Simple, Stored
PROPERTIES	property_id	Simple, Stored
PROPERTIES	square_feet	Simple, Stored
PROPERTIES	owner_name	Simple, Stored
PROPERTIES	owner_contact	Simple, Stored
PROPERTIES	location	Composite, Stored
PROPERTIES	status	Simple, Stored
PROPERTIES	value	Simple, Stored
PROPERTIES	eligible_tenant_type	Multivalued, Stored
ENQUIRIES	enquiry_id	Simple, Stored
ENQUIRIES	enquiry_date	Simple, Stored
ENQUIRIES	status	Simple, Stored
ENQUIRIES	notes	Simple, Stored
DEALS	deal_id	Simple, Stored
DEALS	deal_type	Simple, Stored

DEALS	deal_date	Simple, Stored
DEALS	deal_value	Simple, Stored
DEALS	deal_status	Simple, Stored
DEALS	rental_start_date	Simple, Stored
DEALS	rental_end_date	Simple, Stored
DEALS	broker_earning	Simple, Derived
DEALS	company_earning	Simple, Derived
COMMISSION	commission_id	Simple, Stored
COMMISSION	owner_brokerage_percent	Simple, Stored
COMMISSION	client_brokerage_percent	Simple, Stored
LEASES	lease_id	Simple, Stored
LEASES	lease_type	Simple, Stored
LEASES	security_deposit	Simple, Stored
LEASES	duration_months	Simple, Stored
REVIEWS	review_id	Simple, Stored
REVIEWS	comments	Simple, Stored
REVIEWS	rating	Simple, Stored
REVIEWS	review_date	Simple, Stored

Note: all other attributes are simple and stored. location is composite (area, city, pincode) but is kept as one text column. eligible_tenant_type is multivalued because a property can suit more than one tenant type. broker_earning and company_earning are derived because they can be calculated from deal_value and the brokerage percentages in COMMISSION.

5. ER to relational schema conversion

Mapping rules applied

- Each entity becomes a table; its key attribute becomes the primary key.
- Each simple or stored attribute becomes a column. The composite attribute location is kept as one column.
- The multivalued attribute eligible_tenant_type becomes a separate table with the property key.
- Derived attributes (broker_earning, company_earning) are not stored; they are calculated when needed.
- Each 1:N relationship is mapped by placing the primary key of the 1-side as a foreign key in the N-side table.
- The 1:1 relationship (enquiry converts to deal) is mapped with a unique foreign key in DEALS.

Relational schema

```
BROKERS (broker_id PK, name, email, phone, join_date)
COMMISSION (commission_id PK, owner_brokerage_percent, client_brokerage_percent,
broker_id FK -> BROKERS)
PROPERTIES (property_id PK, square_feet, owner_name, owner_contact, location, status, value, broker_id
FK -> BROKERS, commission_id FK -> COMMISSION)
PROPERTY_TENANT_TYPE (property_id FK -> PROPERTIES, tenant_type) PK = (property_id, tenant_type)
LEASES (lease_id PK, lease_type, security_deposit, duration_months,
property_id FK -> PROPERTIES)
CLIENTS (client_id PK, name, email, phone, client_type)
ENQUIRIES (enquiry_id PK, enquiry_date, status, notes, client_id FK -> CLIENTS,
property_id FK -> PROPERTIES, broker_id FK -> BROKERS)
DEALS (deal_id PK, deal_type, deal_date, deal_value, deal_status, rental_start_date, rental_end_date,
client_id FK -> CLIENTS, property_id FK -> PROPERTIES, broker_id FK -> BROKERS, enquiry_id FK
-> ENQUIRIES, UNIQUE)
REVIEWS (review_id PK, comments, rating, review_date, client_id FK -> CLIENTS)
```

6. Normalization

Normalization removes redundancy and update, insert and delete anomalies. Each normal form is checked for the schema above.

First normal form (1NF)

Every column must hold a single atomic value and there must be no repeating groups. The multivalued attribute eligible_tenant_type would put several values in one cell, so it is moved to PROPERTY_TENANT_TYPE, where each row holds one value. Composite location is stored as one atomic text value, and derived attributes are not stored. All tables have a primary key, so the schema is in 1NF.

Second normal form (2NF)

The table must be in 1NF and no non-key column may depend on only part of a composite key. Suppose tenant types were kept with the property in one table with key (property_id, tenant_type): owner_name and value would depend on property_id alone, which is a partial dependency. Separating PROPERTY_TENANT_TYPE from PROPERTIES removes it. All other tables have a single-column key, so partial dependencies cannot occur.

Third normal form (3NF)

The table must be in 2NF and have no transitive dependency (a non-key column depending on another non-key column). The brokerage percentages are kept only in COMMISSION and referenced by commission_id; storing them in PROPERTIES would create property_id -> commission_id -> percentages. In the same way, broker_earning and company_earning depend on deal_value and the percentages rather than on deal_id alone, so they are treated as derived and not stored in DEALS. Owner details are assumed to belong to a single property listing; if one owner could own many properties, an OWNER table would be needed.

Boyce-Codd normal form (BCNF)

For every functional dependency $X \rightarrow Y$, X must be a candidate key. The dependencies of the final schema are:

BROKERS	broker_id -> name, email, phone, join_date
COMMISSION	commission_id -> owner_brokerage_percent, client_brokerage_percent, broker_id
PROPERTIES	property_id -> square_feet, owner_name, owner_contact, location, status, value, broker_id, commission_id
PROPERTY_TENANT_ TYPE	(property_id, tenant_type) -> no other columns (all-key table)
LEASES	lease_id -> lease_type, security_deposit, duration_months, property_id
CLIENTS	client_id -> name, email, phone, client_type
ENQUIRIES	enquiry_id -> enquiry_date, status, notes, client_id, property_id, broker_id
DEALS	deal_id -> deal_type, deal_date, deal_value, deal_status, rental_start_date, rental_end_date, client_id, property_id, broker_id, enquiry_id
REVIEWS	review_id -> comments, rating, review_date, client_id

In every table the only determinant is the primary key (a candidate key), so all tables are in BCNF, and therefore also in 1NF, 2NF and 3NF.